THE MAYER-VIETORIS SEQUENCE FOR ETALE COHOMOLOGY OF A UNION OF TWO OPENS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; the short exactness on injectives and the passage to the long exact sequence are both inside the counted span.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Injective abelian sheaf on the small etale site and the sections of its restriction. Same chapter, injective abelian sheaves and the vanishing of the higher cohomology of their restrictions, which makes the sequence of section complexes short exact. Supplies the short exactness on which the proof rests.
- Long exact cohomology sequence attached to a short exact sequence of complexes. Same chapter together with the chapter Homology, the long exact sequence of a short exact sequence of complexes of abelian groups. Supplies the long exact sequence asserted in the statement.

Source reference. Original lemma 093X: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 7204–7204. Navigation pointer invoked at proof line 7204; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03OR: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 1702–1702. Navigation pointer invoked at proof line 7210; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 03OV: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 1758–1758. Navigation pointer invoked at proof line 7210; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1 (Mayer-Vietoris for étale cohomology)**.** *Let X be a scheme. Suppose that $X = U \cup V$ is a union of two opens. For any abelian sheaf $\mathcal{F}$ on $X_{étale}$ there exists a long exact cohomology sequence*

$$\begin{aligned} 0 \to H^0_{étale}(X, \mathcal{F}) \to H^0_{étale}(U, \mathcal{F}) \oplus H^0_{étale}(V, \mathcal{F}) \to H^0_{étale}(U \cap V, \mathcal{F}) \\ \to H^1_{étale}(X, \mathcal{F}) \to H^1_{étale}(U, \mathcal{F}) \oplus H^1_{étale}(V, \mathcal{F}) \to H^1_{étale}(U \cap V, \mathcal{F}) \to \ldots \end{aligned}$$

This long exact sequence is functorial in $\mathcal{F}$.

Proof. Observe that if $\mathcal{I}$ is an injective abelian sheaf, then

$$0 \to \mathcal{I}(X) \to \mathcal{I}(U) \oplus \mathcal{I}(V) \to \mathcal{I}(U \cap V) \to 0$$

is exact. This is true in the first and middle spots as $\mathcal{I}$ is a sheaf. It is true on the right, because $\mathcal{I}(U) \to \mathcal{I}(U \cap V)$ is surjective by Cohomology on Sites, Lemma 093X. Another way to prove it would be to show that the cokernel of the map $\mathcal{I}(U) \oplus \mathcal{I}(V) \to \mathcal{I}(U \cap V)$ is the first Čech cohomology group of $\mathcal{I}$ with respect to the covering $X = U \cup V$ which vanishes by Lemmas 03OR and 03OV. Thus, if $\mathcal{F} \to \mathcal{I}^\bullet$ is an injective resolution, then

$$0 \to \mathcal{I}^\bullet(X) \to \mathcal{I}^\bullet(U) \oplus \mathcal{I}^\bullet(V) \to \mathcal{I}^\bullet(U \cap V) \to 0$$

is a short exact sequence of complexes and the associated long exact cohomology sequence is the sequence of the statement of the lemma. □